

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE SUPPLEMENATRY SUMMER EXAMINATION – 2024 Course: B. Tech. Class: Final Year Semester: VII Branch : Electrical Engineering / Electrical Engineering (Electronics and Power)/ electrical & Electronics Engg / Electrical & Power Engineering Subject Code & Name: Electric and Hybrid Electric Vehicles BTEEOE705B Max Marks: 60 Date: 10/07/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q.1	Solve Any Two of the following.		12
A)	Compare the performance of EV and HEV?	Remember	6
B)	Explain about the vehicle power source characteristics and Explain about the vehicle transmission characteristics	Understand	6
C)	Under what condition a pure EV can be chosen as a best option compared to hybrid vehicles considering the impact on climate change?	Applying	6
Q.2	Solve Any Two of the following.		12
A)	Explain the different power flow control modes of a typical parallel hybrid system with the help of block diagrams	Applying	6
B)	Draw six different configurations of drivetrains in electric vehicles. Briefly explain each configuration	Remember	6
C)	Explain the fuel efficiency analysis of hybrid drive trains and electric drive trains?	Understand	6
Q.3	Solve Any Two of the following.		12
A)	What are the important subsystems in an electric and hybrid vehicle?	Remember	6
B)	Explain the forward motoring and regenerative (forward) braking control of a dc motor with a single chopper. Give circuit diagram, and show the quadrants of operation	Applying	6
C)	Discuss the possible power converter topologies that can be used in Induction motor drive for EV/HEV.	Applying	6

Q.4	Solve Any Two of the following.		12
A)	What are factors affecting the performance of batteries used in EVs?	Understand	6
B)	Compare different batteries based on their specific energy, specific power and suitability for HV/EHV applications	Remember	6
C)	Describe the terms State-of-Charge and Depth-of-Discharge as applied to batteries.	Understand	6
Q.5	Solve Any Two of the following.		12
A)	What is power follower strategy for energy management in hybrids? Is this a rule based strategy or an optimization based strategy?	Understand	6
B)	Classify and explain the basic principles of rule based energy management system	Remember	6
C)	Draw the block diagram of a general Fuzzy Logic Controller (FLC) and show the core components of the FLC and the inputs and outputs relevant to a hybrid electric vehicle control.	Applying	6
	*** End ***		